Yale AP PhD Interview

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Coarse-grained quantum state tomography with optimal POVM construction

η_0, η_1: the probability of **failing** to detect a signal from the $|0\rangle$ and $|1\rangle$ state.

$$\begin{aligned} \Pi_0 &= \eta_0|0\rangle\langle 0| + \eta_1|1\rangle\langle 1| && \text{no detection} \\ \Pi_1 &= (1-\eta_0)|0\rangle\langle 0| + (1-\eta_1)|1\rangle\langle 1| && \text{a detection event} \end{aligned}$$

Extending this framework to N-qubit system, the signal detection probability becomes:

$$M_{\mathrm{CG}} = I^{\otimes N} - \Pi_0^{\otimes N}$$

resulting in two positive semidefinite observable operators $\{M_{\mathrm{CG}}, I^{\otimes N} - M_{\mathrm{CG}}\}$.

By implementing unitary gates $G^k(\hat{\theta})$ via parameterized quantum circuits, an extended set of CG POVM operators, $\Omega^k = G^{k\dagger}(\hat{\theta}) M_{\mathrm{CG}} G^k(\hat{\theta})$ can be constructed.

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Idea: Uniformly distributed Measurement Operators across Hilbert space.

Uniformness of the set of measurement operators can be represented by von Neumann entropy of Gram matrix

$$S = -\mathrm{Tr}(\Pi' \ln \Pi') = -\sum_i \lambda_i' \ln \lambda_i'$$

where Gram matrix is defined as each element is the inner product of the POVM bases $\Pi_{ij} = \mathrm{Tr}\Omega^i\Omega^j$, and normalized $\Pi' = \Pi/\mathrm{Tr}\Pi$ so that sum of eigenvalues becomes 1.

To maximize von Neumann entropy S, we iteratively update the circuit parameters $\hat{\theta}$. That is, we solved $\arg\max_{\hat{\theta}} S$.

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(a) The CG POVM based QST process for a two-qubit system is illustrated, where quantum state tomography is performed after an arbitrary quantum operation $\mathcal{O}$ and subsequent projection basis gates $G^k(\theta)$.

(b) Higher von Neumann entropy of the CG POVM set corresponds to reduced infidelity, demonstrating robustness against the statistical noise.

(c) CG POVM sets with higher entropy provide more consistent reconstruction results, demonstrating their state-independent performance across a variety of quantum states.

Enhanced electron nuclear control leveraging hyperfine coupling and radio-frequency drive

Dynamic Decoupling (DD) sequences, consisting of multi-pulse control on the NV spin, have been developed to achieve entangling gates with selected target nuclear spins while reducing decoherence from the broader spin bath.

An additional selective phase-controlled **radio-frequency (rf)** driving on nuclear spins, referred as DDrf gate, enables elaborated control of individual spins.

$$\mathcal{H} = \underbrace{\gamma_c B_z I_z^i + A_{||}^i S_z I_z^i + A_\perp S_z I_x^i}_{\text{Drift Hamiltonian}} + \underbrace{\frac{1}{\sqrt{2}} \Omega_{\mathrm{MW}} S_x}_{\text{NV electron spin control}} + \underbrace{\frac{1}{2} \Omega_{\mathrm{rf}} I_x}_{^{13}\text{C spin control}}$$

$$\rightarrow \mathcal{H} = |0\rangle\langle 0| \otimes \omega_0 I_z + |-1\rangle\langle -1| \otimes \left(\omega_L I_z - A_{||} I_z - A_\perp I_x\right)$$

$$\rightarrow \mathcal{H} = |0\rangle\langle 0| \otimes H_0 + |-1\rangle\langle -1| \otimes H_1$$